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Contents

Intellectual Property Rights 6

Foreword 6

1 Scope 7
1.1 References 7
1.2 Abbreviations 7

2 General 8
2.1 General organization 8
2.2 Naming Convention 10

3 Traffic Channels (TCH) 10
3.1 Speech channel at full rate (TCH/FS and TCH/EFS) 11
3.1.1 Preliminary channel coding for EFR only 11
3.1.1.1 CRC calculation 11
3.1.1.2 Repetition bits 11
3.1.1.3 Correspondence between input and output of preliminary channel coding 12
3.1.2 Channel coding for FR and EFR 12
3.1.2.1 Parity and tailing for a speech frame 12
3.1.2.2 Convolutional encoder 13
3.1.3 Interleaving 13
3.1.4 Mapping on a Burst 13
3.2 Speech channel at half rate (TCH/HS) 13
3.2.1 Parity and tailing for a speech frame 14
3.2.2 Convolutional encoder 14
3.2.3 Interleaving 15
3.2.4 Mapping on a burst 15
3.3 Data channel at full rate, 12.0 kbit/s radio interface rate (9.6 kbit/s services (TCH/F9.6)) 15
3.3.1 Interface with user unit 15
3.3.2 Block code 16
3.3.3 Convolutional encoder 16
3.3.4 Interleaving 16
3.3.5 Mapping on a Burst 16
3.4 Data channel at full rate, 6.0 kbit/s radio interface rate (4.8 kbit/s services (TCH/F4.8)) 16
3.4.1 Interface with user unit 17
3.4.2 Block code 17
3.4.3 Convolutional encoder 17
3.4.4 Interleaving 17
3.4.5 Mapping on a Burst 17
3.5 Data channel at half rate, 6.0 kbit/s radio interface rate (4.8 kbit/s services (TCH/H4.8)) 17
3.5.1 Interface with user unit 18
3.5.2 Block code 18
3.5.3 Convolutional encoder 18
3.5.4 Interleaving 18
3.5.5 Mapping on a Burst 18
3.6 Data channel at full rate, 3.6 kbit/s radio interface rate (2.4 kbit/s and less services (TCH/F2.4)) 18
3.6.1 Interface with user unit 18
3.6.2 Block code 18
3.6.3 Convolutional encoder 18
3.6.4 Interleaving 19
3.6.5 Mapping on a Burst 19
3.7 Data channel at half rate, 3.6 kbit/s radio interface rate (2.4 kbit/s and less services (TCH/H2.4)) 19
3.7.1 Interface with user unit 19
3.7.2 Block code 19
3.7.3 Convolutional encoder 19
3.7.4 Interleaving 19
3.7.5 Mapping on a Burst 19

3.8 Data channel at full rate, 14.5 kbit/s radio interface rate (14.4 kbit/s services (TCH/F14.4)) 20
3.8.1 Interface with user unit 20
3.8.2 Block code 20
3.8.3 Convolutional encoder 20
3.8.4 Interleaving 20
3.8.5 Mapping on a Burst 20

4 Control Channels 21
4.1 Slow associated control channel (SACCH) 21
4.1.1 Block constitution 21
4.1.2 Block code 21
4.1.3 Convolutional encoder 21
4.1.4 Interleaving 22
4.1.5 Mapping on a Burst 22
4.2 Fast associated control channel at full rate (FACCH/F) 22
4.2.1 Block constitution 22
4.2.2 Block code 22
4.2.3 Convolutional encoder 22
4.2.4 Interleaving 22
4.2.5 Mapping on a Burst 22
4.3 Fast associated control channel at half rate (FACCH/H) 23
4.3.1 Block constitution 23
4.3.2 Block code 23
4.3.3 Convolutional encoder 23
4.3.4 Interleaving 23
4.3.5 Mapping on a Burst 24
4.4 Broadcast control, Paging, Access grant, Notification and Cell broadcast channels (BCCH, PCH, AGCH, NCH, CBCH) 24
4.5 Stand-alone dedicated control channel (SDCCH) 24
4.6 Random access channel (RACH) 25
4.7 Synchronization channel (SCH) 25
4.8 Access Burst on circuit switched channels other than RACH 26
4.9 Access Bursts for uplink access on a channel used for VGCS 26

5 Packet Switched Channels 26
5.1 Packet data traffic channel (PDTCH) 26
5.1.1 Packet data block type 1 (CS-1) 26
5.1.2 Packet data block type 2 (CS-2) 26
5.1.2.1 Block constitution 26
5.1.2.2 Block code 26
5.1.2.3 Convolutional encoder 27
5.1.2.4 Interleaving 27
5.1.2.5 Mapping on a burst 27
5.1.3 Packet data block type 3 (CS-3) 28
5.1.3.1 Block constitution 28
5.1.3.2 Block code 28
5.1.3.3 Convolutional encoder 28
5.1.3.4 Interleaving 29
5.1.3.5 Mapping on a burst 29
5.1.4 Packet data block type 4 (CS-4) 29
5.1.4.1 Block constitution 29
5.1.4.2 Block code 29
5.1.4.3 Convolutional encoder 29
5.1.4.4 Interleaving 30
5.1.4.5 Mapping on a burst 30
5.2 Packet control channels (PACCH, PBCCH, PAGCH, PPCH, PNCH, PTCCH) 30
5.3 Packet random access channel (PRACH) 30
5.3.1 Packet Access Burst 30
5.3.2 Extended Packet Access Burst 30
5.4 Access Burst on packet switched channels other than PRACH 31

Annex A (informative): Summary of Channel Types 43

Annex B (informative): Summary of Polynomials Used for Convolutional Codes.....44

Annex C (informative): Change Request History 45

History 46

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Foreword

This European Standard (Telecommunications series) has been produced by the Special Mobile Group (SMG).

The present document specifies the data blocks given to the encryption unit. It includes the specification of encoding, reordering, interleaving and the stealing flag within the digital cellular telecommunications system (Phase 2+).

The contents of the present document are subject to continuing work within SMG and may change following formal SMG approval. Should SMG modify the contents of the present document it will then be republished by ETSI with an identifying change of release date and an increase in version number as follows:

Version 6.x.y

where:

6 indicates release 1997 of GSM Phase 2+

x the second digit is incremented for changes of substance, i.e. technical enhancements, corrections, updates, etc

y the third digit is incremented when editorial only changes have been incorporated in the specification.

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1 Scope

A reference configuration of the transmission chain is shown in GSM 05.01 [4]. According to this reference configuration, the present document specifies the data blocks given to the encryption unit.

It includes the specification of encoding, reordering, interleaving and the stealing flag. It does not specify the channel decoding method.

The definition is given for each kind of logical channel, starting from the data provided to the channel encoder by the speech coder, the data terminal equipment, or the controller of the Mobile Station (MS) or Base Transceiver Station (BTS). The definitions of the logical channel types used in the present document are given in GSM 05.02 [5], a summary is in annex A.

1.1 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] GSM 01.04: "Digital cellular telecommunications system (Phase 2+); Abbreviations and acronyms".

[2] GSM 04.08: "Digital cellular telecommunications system (Phase 2+); Mobile radio interface layer 3 specification".

[3] GSM 04.21: "Digital cellular telecommunications system (Phase 2+); Rate adaption on the Mobile Station - Base Station System (MS - BSS) interface".

[4] GSM 05.01: "Digital cellular telecommunications system (Phase 2+); Physical layer on the radio path General description".

[5] GSM 05.02: "Digital cellular telecommunications system (Phase 2+); Multiplexing and multiple access on the radio path".

[6] GSM 05.05: "Digital cellular telecommunications system (Phase 2+); Radio Transmission and Reception".

[7] GSM 06.10: "Digital cellular telecommunications system; Full rate speech transcoding".

[8] GSM 06.20: "Digital cellular telecommunications system; Half rate speech transcoding".

[9] GSM 06.60: "Digital cellular telecommunications system; Enhanced Full Rate (EFR) speech transcoding".

[10] GSM 03.64: "Digital cellular telecommunications system (Phase 2+); General Packet Radio Service (GPRS); Overall description of the GPRS Radio Interface; Stage 2".

1.2 Abbreviations

Abbreviations used in the present document are listed in GSM 01.04.

2 General

2.1 General organization

Each channel has its own coding and interleaving scheme. However, the channel coding and interleaving is organized in such a way as to allow, as much as possible, a unified decoder structure.

Each channel uses the following sequence and order of operations:

- The information bits are coded with a systematic block code, building words of information + parity bits.
- These information + parity bits are encoded with a convolutional code, building the coded bits.
- Reordering and interleaving the coded bits, and adding a stealing flag, gives the interleaved bits.

All these operations are made block by block, the size of which depends on the channel. However, most of the channels use a block of 456 coded bits which is interleaved and mapped onto bursts in a very similar way for all of them. Figure 1 gives a diagram showing the general structure of the channel coding.

This block of 456 coded bits is the basic structure of the channel coding scheme. In the case of full rate speech TCH, this block carries the information of one speech frame. In case of control channels, it carries one message.

In the case of half rate speech TCH, the information of one speech frame is carried in a block of 228 coded bits.

In the case of the Enhanced full rate speech the information bits coming out of the source codec first go though a preliminary channel coding. then the channel coding as described above takes place.

In the case of a packet switched channel the block of 456 coded bits carries one radio block.

In the case of FACCH, a coded message block of 456 bits is divided into eight sub-blocks. The first four sub-blocks are sent by stealing the even numbered bits of four timeslots in consecutive frames used for the TCH. The other four sub-blocks are sent by stealing the odd numbered bits of the relevant timeslot in four consecutive used frames delayed 2 or 4 frames relative to the first frame. Along with each block of 456 coded bits there is, in addition, a stealing flag (8 bits), indicating whether the block belongs to the TCH or to the FACCH. In the case of SACCH, BCCH or CCCH, this stealing flag is dummy. In the case of a packet switched channel, these bits are used to indicate the coding scheme used.

Some cases do not fit in the general organization, and use short blocks of coded bits which are sent completely in one timeslot. They are the random access messages of

- the RACH
- or PRACH

on uplink and the synchronization information broadcast of the SCH on downlink.

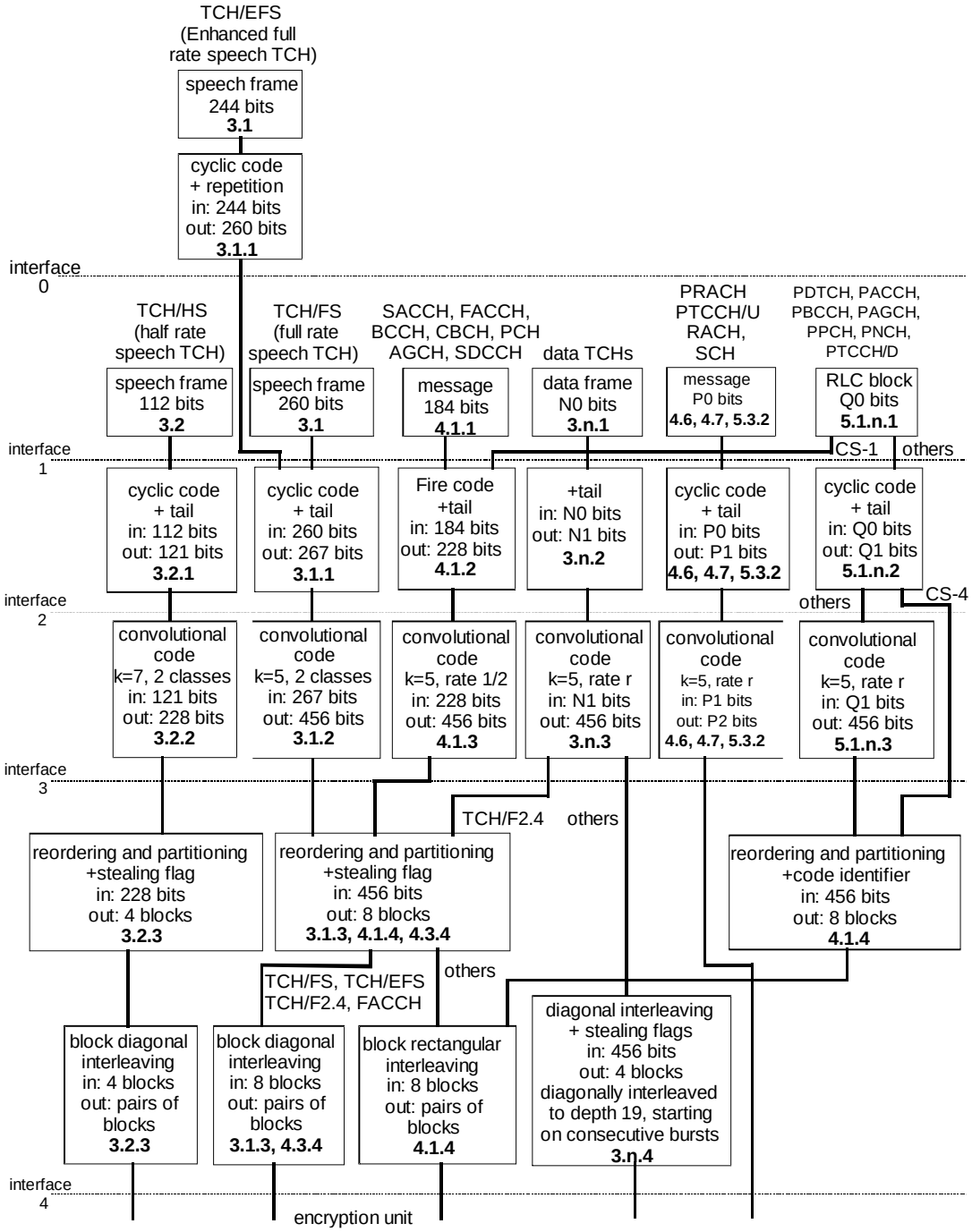


Figure 1: Channel Coding and Interleaving Organization

In each box, the last line indicates the chapter defining the function. In the case of RACH, P0 = 8 and P1 = 18; in the case of SCH, P0 = 25 and P1 = 39. In the case of data TCHs, N0, N1 and n depend on the type of data TCH.

Interfaces:

1) information bits (d);

2) information + parity + tail bits (u);

3) coded bits (c);

4) interleaved bits (e).

2.2 Naming Convention

For ease of understanding a naming convention for bits is given for use throughout the technical specification:

- General naming:

  "k" and "j" for numbering of bits in data blocks and bursts;

  "K_x" gives the amount of bits in one block, where "x" refers to the data type;

  "n" is used for numbering of delivered data blocks where;

  "N" marks a certain data block;

  "B" is used for numbering of bursts or blocks where;

  "B_0" marks the first burst or block carrying bits from the data block with n = 0 (first data block in the transmission).

- Data delivered to the preliminary channel encoding unit (for EFR only):

  s(k) for k = 1..., K_s

- Data delivered by the preliminary channel encoding unit (for EFR only) before bits rearrangement

  w(k) for k = 1..., K_w

- Data delivered to the encoding unit (interface 1 in figure 1):

  d(k) for k = 0,1,...,K_d-1

- Code identifying the used coding scheme (for packet switched channels only):

  q(k) for k = 0,1,..., 7

- Data after the first encoding step (block code, cyclic code; interface 2 in figure 1):

  u(k) for k = 0,1,...,K_u-1

- Data after the second encoding step (convolutional code ; interface 3 in figure 1):

  c(n,k) or c(k) for k = 0,1,...,K_c-1

  n = 0,1,...,N,N+1,...

- Interleaved data:

  i(B,k) for k = 0,1,...,K_i-1

  B = B_0, B_0+1,....

- Bits in one burst (interface 4 in figure 1):

  e(B,k) for k = 0,1,114,115

  B = B_0,B_0+1,...

3 Traffic Channels (TCH)

Two kinds of traffic channel are considered: speech and data. Both of them use the same general structure (see figure 1), and in both cases, a piece of information can be stolen by the FACCH.

3.1 Speech channel at full rate (TCH/FS and TCH/EFS)

The speech coder (whether Full rate or Enhanced full rate) delivers to the channel encoder a sequence of blocks of data. In case of a full rate and enhanced full rate speech TCH, one block of data corresponds to one speech frame.

For the full rate coder each block contains 260 information bits, including 182 bits of class 1 (protected bits), and 78 bits of class 2 (no protection), (see table 2).

The bits delivered by the speech coder are received in the order indicated in GSM 06.10 and have to be rearranged according to table 2 before channel coding as defined in subclauses 3.1.1 to 3.1.4. The rearranged bits are labelled {d(0),d(1),...,d(259)}, defined in the order of decreasing importance.

For the EFR coder each block contains 244 information bits. The block of 244 information bits, labelled s(1).., s(244), passes through a preliminary stage, applied only to EFR (see figure 1) which produces 260 bits corresponding to the 244 input bits and 16 redundancy bits. Those 16 redundancy bits correspond to 8 CRC bits and 8 repetition bits, as described in subclause 3.1.1. The 260 bits, labelled w(1)..w(260), have to be rearranged according to table 7 before they are delivered to the channel encoding unit which is identical to that of the TCH/FS. The 260 bits block includes 182 bits of class 1(protected bits) and 78 bits of class 2 (no protection). The class 1 bits are further divided into the class 1a and class 1b, class 1a bits being protected by a cyclic code and the convolutional code whereas the class 1b are protected by the convolutional code only.

3.1.1 Preliminary channel coding for EFR only

3.1.1.1 CRC calculation

An 8-bit CRC is used for error-detection. These 8 parity bits (bits w253-w260) are generated by the cyclic generator polynomial: $g(D) = D^8 + D^4 + D^3 + D^2 + 1$ from the 65 most important bits (50 bits of class 1a and 15 bits of class 1b). These 65 bits *(b(1)-b(65))* are taken from the table 5 in the following order (read row by row, left to right):

s39	s40	s41	s42	s43	s44	s48	s87	s45	s2
s3	s8	s10	s18	s19	s24	s46	s47	s142	s143
s144	s145	s146	s147	s92	s93	s195	s196	s98	s137
s148	s94	s197	s149	s150	s95	s198	s4	s5	s11
s12	s16	s9	s6	s7	s13	s17	s20	s96	s199
s1	s14	s15	s21	s25	s26	s28	s151	s201	s190
s240	s88	s138	s191	s241					

The encoding is performed in a systematic form, which means that, in GF(2), the polynomial:

$b(1)D^{72} + b(2)D^{71} + ... + b(65)D^8 + p(1)D^7 + p(2)D^6 + ... + p(7)D^1 + p(8)$

p(1) - p(8): the parity bits (w253-w260)

b(1) - b(65) = the data bits from the table above

when divided by *g(D),* yields a remainder equal to 0.

3.1.1.2 Repetition bits

The repeated bits are s70, s120, s173 and s223. They correspond to one of the bits in each of the PULSE_5, the most significant one not protected by the channel coding stage.

3.1.1.3 Correspondence between input and output of preliminary channel coding

The preliminary coded bits w(k) for k = 1 to 260 are hence defined by:

w(k) = s(k) for k = 1 to 71

w(k) = s(k-2) for k = 74 to 123

w(k) = s(k-4) for k = 126 to 178

w(k) = s(k-6) for k = 181 to s230

w(k) = s(k-8) for k = 233 to s252

Repetition bits:

w(k) = s(70) for k = 72 and 73

w(k) = s(120) for k = 124 and 125

w(k) = s(173) for k = 179 and 180

w(k) = s(223) for k = 231 and 232

Parity bits:

w(k = p(k-252) for k = 253 to 260

3.1.2 Channel coding for FR and EFR

3.1.2.1 Parity and tailing for a speech frame

a) Parity bits:

The first 50 bits of class 1 (**known as class 1a for the EFR)** are protected by three parity bits used for error detection. These parity bits are added to the 50 bits, according to a degenerate (shortened) cyclic code (53,50,2), using the generator polynomial:

$g(D) = D^3 + D + 1$

The encoding of the cyclic code is performed in a systematic form, which means that, in GF(2), the polynomial:

$d(0)D^{52} + d(1)D^{51} + ... + d(49)D^3 + p(0)D^2 + p(1)D + p(2)$

where p(0), p(1), p(2) are the parity bits, when divided by g(D), yields a remainder equal to:

$1 + D + D^2$

b) Tailing bits and reordering:

The information and parity bits of class 1 are reordered, defining 189 information + parity + tail bits of class 1, {u(0),u(1),...,u(188)} defined by:

u(k) = d(2k) and u(184-k) = d(2k+1) for k = 0,1,...,90

u(91+k) = p(k) for k = 0,1,2

u(k) = 0 for k = 185,186,187,188 (tail bits)

3.1.2.2 Convolutional encoder

The class 1 bits are encoded with the 1/2 rate convolutional code defined by the polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

The coded bits {c(0), c(1),..., c(455)} are then defined by:

- class 1: $c(2k) = u(k) + u(k-3) + u(k-4)$
  $c(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4)$ for $k = 0,1,...,188$
  $u(k) = 0$ for $k < 0$

- class 2: $c(378+k) = d(182+k)$ for $k = 0,1,....,77$

3.1.3 Interleaving

The coded bits are reordered and interleaved according to the following rule:

$i(B,j) = c(n,k)$, for $k = 0,1,...,455$

$n = 0,1,...,N,N+1,...$

$B = B_0 + 4n + (k \bmod 8)$

$j = 2((49k) \bmod 57) + ((k \bmod 8) \operatorname{div} 4)$

See table 1. The result of the interleaving is a distribution of the reordered 456 bits of a given data block, n = N, over 8 blocks using the even numbered bits of the first 4 blocks ($B = B_0 + 4N + 0, 1, 2, 3$) and odd numbered bits of the last 4 blocks ($B = B_0 + 4N + 4, 5, 6, 7$). The reordered bits of the following data block, n = N+1, use the even numbered bits of the blocks $B = B_0 + 4N + 4, 5, 6, 7$ ($B = B_0 + 4(N+1) + 0, 1, 2, 3$) and the odd numbered bits of the blocks $B = B_0 + 4(N+1) + 4, 5, 6, 7$. Continuing with the next data blocks shows that one block always carries 57 bits of data from one data block (n = N) and 57 bits of data from the next block (n = N+1), where the bits from the data block with the higher number always are the even numbered data bits, and those of the data block with the lower number are the odd numbered bits.

The block of coded data is interleaved "block diagonal", where a new data block starts every 4th block and is distributed over 8 blocks.

3.1.4 Mapping on a Burst

The mapping is given by the rule:

$e(B,j) = i(B,j)$ and $e(B,59+j) = i(B,57+j)$ for $j = 0,1,...,56$

and

$e(B,57) = hl(B)$ and $e(B,58) = hu(B)$

The two bits, labelled hl(B) and hu(B) on burst number B are flags used for indication of control channel signalling. For each TCH/FS block not stolen for signalling purposes:

$hu(B) = 0$ for the first 4 bursts (indicating status of even numbered bits)

$hl(B) = 0$ for the last 4 bursts (indicating status of odd numbered bits)

For the use of hl(B) and hu(B) when a speech frame is stolen for signalling purposes see subclause 4.2.5.

3.2 Speech channel at half rate (TCH/HS)

The speech coder delivers to the channel encoder a sequence of blocks of data. In case of a half rate speech TCH, one block of data corresponds to one speech frame. Each block contains 112 bits, including 95 bits of class 1 (protected bits), and 17 bits of class 2 (no protection), see tables 3a and 3b.

The bits delivered by the speech coder are received in the order indicated in GSM 06.20 and have to be arranged according to either table 3a or table 3b before channel encoding as defined in subclauses 3.2.1 to 3.2.4. The rearranged bits are labelled {d(0),d(1),...,d(111)}. Table 3a has to be taken if parameter Mode = 0 (which means that the speech encoder is in unvoiced mode), while table 3b has to be taken if parameter Mode = 1, 2 or 3 (which means that the speech encoder is in voiced mode).

3.2.1 Parity and tailing for a speech frame

a) Parity bits:

The most significant 22 class 1 bits d(73),d(74),...,d(94) are protected by three parity bits used for error detection. These bits are added to the 22 bits, according to a cyclic code using the generator polynomial:

$g(D) = D^3 + D + 1$

The encoding of the cyclic code is performed in a systematic form, which means that, in GF(2), the polynomial:

$d(73)D^{24} + d(74)D^{23} + ... + d(94)D^3 + p(0)D^2 + p(1)D + p(2)$

where p(0), p(1), p(2) are the parity bits, when divided by g(D), yields a remainder equal to:

$1 + D + D^2$.

b) Tail bits and reordering:

The information and parity bits of class 1 are reordered, defining 104 information + parity + tail bits of class 1, {u(0),u(1),...,u(103)} defined by:

u(k) = d(k)	for k = 0,1,...,94
u(k) = p(k-95)	for k = 95,96,97
u(k) = 0	for k = 98,99,...,103 (tail bits)

3.2.2 Convolutional encoder

The class 1 bits are encoded with the punctured convolutional code defined by the mother polynomials:

$G4 = 1 + D^2 + D^3 + D^5 + D^6$

$G5 = 1 + D + D^4 + D^6$

$G6 = 1 + D + D^2 + D^3 + D^4 + D^6$

and the puncturing matrices:

(1,0,1) for {u(0),u(1),...,u(94)} (class 1 information bits);

and {u(98),u(99),...,u(103)} (tail bits).

(1,1,1) for {u(95),u(96),u(97)} (parity bits)

In the puncturing matrices, a 1 indicates no puncture and a 0 indicates a puncture.

The coded bits {c(0),c(1),...,c(227)} are then defined by:

class 1 information bits:

c(2k) = u(k)+u(k-2)+u(k-3)+ (k-5)+u(k-6)
c(2k+1) = u(k)+u(k-1)+u(k-2)+u(k-3)+u(k-4)+u(k-6) for k = 0,1,...,94;u(k) = 0 for k<0

parity bits:

c(3k-95) = u(k)+u(k-2)+u(k-3)+u(k-5)+u(k-6)
c(3k-94) = u(k)+u(k-1)+u(k-4)+u(k-6)
c(3k-93) = u(k)+u(k-1)+u(k-2)+u(k-3)+u(k-4)+u(k-6) for k = 95,96,97

tail bits:

c(2k+3) = u(k)+u(k-2)+u(k-3)+u(k-5)+u(k-6)
c(2k+4) = u(k)+u(k-1)+u(k-2)+u(k-3)+u(k-4)+u(k-6) for k = 98,99,...,103

class 2 information bits:

c(k+211) = d(k+95) for k = 0,1,...,16

3.2.3 Interleaving

The coded bits are reordered and interleaved according to the following rule:

i(B,j) = c(n,k) for k = 0,1,...,227
n = 0,1,...,N,N+1,...
B = B0 + 2n + b

The values of b and j in dependence of k are given by table 4.

The result of the interleaving is a distribution of the reordered 228 bits of a given data block, n = N, over 4 blocks using the even numbered bits of the first 2 blocks (B = B0+2N+0,1) and the odd numbered bits of the last 2 blocks (B = B0+2N+2,3). The reordered bits of the following data block, n = N + 1, use the even numbered bits of the blocks B = B0 + 2N + 2,3 (B = B0+2(N+1)+0,1) and the odd numbered bits of the blocks B = B0 + 2(N+1) + 2,3. Continuing with the next data blocks shows that one block always carries 57 bits of data from one data block (n = N) and 57 bits from the next block (n = N+1), where the bits from the data block with the higher number always are the even numbered data bits, and those of the data block with the lower number are the odd numbered bits. The block of coded data is interleaved "block diagonal", where a new data block starts every 2nd block and is distributed over 4 blocks.

3.2.4 Mapping on a burst

The mapping is given by the rule:

e(B,j) = i(B,j) and e(B,59+j) = i(B,57+j) for j = 0,1,...,56

and

e(B,57) = hl(B) and e(B,58) = hu(B)

The two bits, labelled hl(B) and hu(B) on burst number B are flags used for indication of control channel signalling. For each TCH/HS block not stolen for signalling purposes:

hu(B) = 0 for the first 2 bursts (indicating status of the even numbered bits)

hl(B) = 0 for the last 2 bursts (indicating status of the odd numbered bits)

For the use of hl(B) and hu(B) when a speech frame is stolen for signalling purposes, see subclause 4.3.5.

3.3 Data channel at full rate, 12.0 kbit/s radio interface rate (9.6 kbit/s services (TCH/F9.6))

The definition of a 12.0 kbit/s radio interface rate data flow for data services is given in GSM 04.21.

3.3.1 Interface with user unit

The user unit delivers to the encoder a bit stream organized in blocks of 60 information bits (data frames) every 5 ms. Four such blocks are dealt with together in the coding process {d(0),...,d(239)}. For non-transparent services those four blocks shall align with one 240-bit RLP frame.

3.3.2 Block code

The block of 4 * 60 information bits is not encoded, but only increased with 4 tail bits equal to 0 at the end of the block.

$u(k) = d(k)$ for k = 0,1,...,239

$u(k) = 0$ for k = 240,241,242,243 (tail bits)

3.3.3 Convolutional encoder

This block of 244 bits {u(0),...,u(243)} is encoded with the 1/2 rate convolutional code defined by the following polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

resulting in 488 coded bits {C(0), C(1),..., C(487)} with

$C(2k) = u(k) + u(k-3) + u(k-4)$

$C(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4)$ for k = 0,1,...,243 ; u(k) = 0 for k < 0

The code is punctured in such a way that the following 32 coded bits:

{C(11+15j) for j = 0,1,...,31} are not transmitted.

The result is a block of 456 coded bits, {c(0),c(1),..., c(455)}

3.3.4 Interleaving

The coded bits are reordered and interleaved according to the following rule:

$i(B,j) = c(n,k)$ for k = 0,1,...,455
n = 0,1,...,N,N + 1,...
$B = B_0 + 4n + (k \bmod 19) + (k \operatorname{div} 114)$
$j = (k \bmod 19) + 19 (k \bmod 6)$

The result of the interleaving is a distribution of the reordered 114 bit of a given data block, n = N, over 19 blocks, 6 bits equally distributed in each block, in a diagonal way over consecutive blocks.

Or in other words the interleaving is a distribution of the encoded, reordered 456 bits from four given input data blocks, which taken together give n = N, over 22 bursts, 6 bits equally distributed in the first and 22nd bursts, 12 bits distributed in the second and 21st bursts, 18 bits distributed in the third and 20th bursts and 24 bits distributed in the other 16 bursts.

The block of coded data is interleaved "diagonal", where a new block of coded data starts with every fourth burst and is distributed over 22 bursts.

3.3.5 Mapping on a Burst

The mapping is done as specified for TCH/FS in subclause 3.1.4. On bitstealing by a FACCH, see subclause 4.2.5.

3.4 Data channel at full rate, 6.0 kbit/s radio interface rate (4.8 kbit/s services (TCH/F4.8))

The definition of a 6.0 kbit/s radio interface rate data flow for data services is given in GSM 04.21.

3.4.1 Interface with user unit

The user unit delivers to the encoder a bit stream organized in blocks of 60 information bits (data frames) every 10 ms, {d(0),d(1),...,d(59)}.

In the case where the user unit delivers to the encoder a bit stream organized in blocks of 240 information bits every 40 ms (e.g. RLP frames), the bits {d(0),d(1),...,d(59),d(60),...,d(60+59), d(2*60),...,d(2*60+59), d(3*60),...,d(3*60+59)} shall be treated as four blocks of 60 bits each as described in the remainder of this clause. To ensure end-to-end synchronization of the 240 bit blocks, the resulting block after coding of the first 120 bits {d(0),d(1),...,d(60+59)} shall be transmitted in one of the transmission blocks B0, B2, B4 of the channel mapping defined in GSM 05.02.

3.4.2 Block code

Sixteen bits equal to 0 are added to the 60 information bits, the result being a block of 76 bits, {u(0),u(1),...,u(75)}, with:

$u(19k+p) = d(15k+p)$ for k = 0,1,2,3 and p = 0,1,...,14;

$u(19k+p) = 0$ for k = 0,1,2,3 and p = 15,16,17,18.

Two such blocks forming a block of 152 bits {u'(0),u'(1),...,u'(151)} are dealt with together in the rest of the coding process:

$u'(k) = u1(k)$, k = 0,1,...,75 (u1 = 1st block)

$u'(k+76) = u2(k)$, k = 0,1,...,75 (u2 = 2nd block)

3.4.3 Convolutional encoder

This block of 152 bits is encoded with the convolutional code of rate 1/3 defined by the following polynomials:

$G1 = 1 + D + D^3 + D^4$

$G2 = 1 + D^2 + D^4$

$G3 = 1 + D + D^2 + D^3 + D^4$

The result is a block of 3 * 152 = 456 coded bits, {c(0),c(1),...,c(455)}:

$c(3k) = u'(k) + u'(k-1) + u'(k-3) + u'(k-4)$

$c(3k+1) = u'(k) + u'(k-2) + u'(k-4)$

$c(3k+2) = u'(k) + u'(k-1) + u'(k-2) + u'(k-3) + u'(k-4)$ for k = 0,1,...,151;

$u'(k) = 0$ for $k < 0$

3.4.4 Interleaving

The interleaving is done as specified for the TCH/F9.6 in subclause 3.3.4.

3.4.5 Mapping on a Burst

The mapping is done as specified for the TCH/FS in subclause 3.1.4. On bitstealing for signalling purposes by a FACCH, see subclause 4.2.5.

3.5 Data channel at half rate, 6.0 kbit/s radio interface rate (4.8 kbit/s services (TCH/H4.8))

The definition of a 6.0 kbit/s radio interface rate data flow for data services is given in GSM 04.21.

3.5.1 Interface with user unit

The user unit delivers to the encoder a bit stream organized in blocks of 60 information bits (data frames) every 10 ms. Four such blocks are dealt with together in the coding process, {d(0),d(1),...,d(239)}.

For non-transparent services those four blocks shall align with one complete 240-bit RLP frame.

3.5.2 Block code

The block encoding is done as specified for the TCH/F9.6 in subclause 3.3.2.

3.5.3 Convolutional encoder

The convolutional encoding is done as specified for the TCH/F9.6 in subclause 3.3.3.

3.5.4 Interleaving

The interleaving is done as specified for the TCH/F9.6 in subclause 3.3.4.

3.5.5 Mapping on a Burst

The mapping is done as specified for the TCH/FS in subclause 3.1.4. On bitstealing for signalling purposes by a FACCH, see subclause 4.3.5.

3.6 Data channel at full rate, 3.6 kbit/s radio interface rate (2.4 kbit/s and less services (TCH/F2.4))

The definition of a 3.6 kbit/s radio interface rate data flow for data services is given in GSM 04.21.

3.6.1 Interface with user unit

The user unit delivers to the encoder a bit stream organized in blocks of 36 information bits (data frames) every 10 ms. Two such blocks are dealt with together in the coding process, {d(0),d(1),...,d(71)}.

3.6.2 Block code

This block of 72 information bits is not encoded, but only increased with four tail bits equal to 0 at the end of the block.

$u(k) = d(k), \quad k = 0,1,...,71$

$u(k) = 0 \quad , \quad k = 72,73,74,75$ (tail bits);

3.6.3 Convolutional encoder

This block of 76 bits {u(0),u(1),...,u(75)} is encoded with the convolutional code of rate 1/6 defined by the following polynomials:

$G1 = 1 + D + D^3 + D^4$

$G2 = 1 + D^2 + D^4$

$G3 = 1 + D + D^2 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

$G2 = 1 + D^2 + D^4$

$G3 = 1 + D + D^2 + D^3 + D^4$

The result is a block of 456 coded bits:

{c(0), c(1),...,c(455)}, defined by

c(6k) = c(6k+3) = u(k) + u(k-1) + u(k-3) + u(k-4)

c(6k+1) = c(6k+4) = u(k) + u(k-2) + u(k-4)

c(6k+2) = c(6k+5) = u(k) + u(k-1) + u(k-2) + u(k-3) + u(k-4), for k = 0,1,...,75;

u(k) = 0 for k < 0

3.6.4 Interleaving

The interleaving is done as specified for the TCH/FS in subclause 3.1.3.

3.6.5 Mapping on a Burst

The mapping is done as specified for the TCH/FS in subclause 3.1.4.

3.7 Data channel at half rate, 3.6 kbit/s radio interface rate (2.4 kbit/s and less services (TCH/H2.4))

The definition of a 3.6 kbit/s radio interface rate data flow for data services is given in GSM 04.21.

3.7.1 Interface with user unit

The user unit delivers to the encoder a bit stream organized in blocks of 36 information bits (data frames) every 10 ms. Two such blocks are dealt with together in the coding process, {d(0),d(1),...,d(71)}.

3.7.2 Block code

The block of 72 information bits is not encoded, but only increased with 4 tail bits equal to 0, at the end of the block.

Two such blocks forming a block of 152 bits {u(0),u(1),...,u(151)} are dealt with together in the rest of the coding process.

u(k) = d1(k), k = 0,1,...,75 (d1 = 1st information block)

u(k+76) = d2(k), k = 0,1,...,75 (d2 = 2nd information block)

u(k) = 0, k = 72,73,74,75,148,149,150,151 (tail bits)

3.7.3 Convolutional encoder

The convolutional encoding is done as specified for the TCH/F4.8 in subclause 3.4.3.

3.7.4 Interleaving

The interleaving is done as specified for the TCH/F9.6 in subclause 3.3.4.

3.7.5 Mapping on a Burst

The mapping is done as specified for the TCH/FS in subclause 3.1.4. On bit stealing for signalling purposes by a FACCH, see subclause 4.3.5.

3.8 Data channel at full rate, 14.5 kbit/s radio interface rate (14.4 kbit/s services (TCH/F14.4))

The definition of a 14.5 kbit/s radio interface rate data flow for data services is given in GSM 04.21.

3.8.1 Interface with user unit

The user unit delivers to the encoder a bit stream organized in blocks of 290 information bits (data frames) every 20 ms.

3.8.2 Block code

The block of 290 information bits is not encoded, but only increased with 4 tail bits equal to 0 at the end of the block.

$u(k) = d(k)$ for $k = 0,1,...,289$

$u(k) = 0$ for $k = 290,291,292,293$ (tail bits)

3.8.3 Convolutional encoder

This block of 294 bits {u(0),...,u(293)} is encoded with the 1/2 rate convolutional code defined by the following polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

resulting in 588 coded bits {C(0), C(1),..., C(587)} with

$C(2k) = u(k) + u(k-3) + u(k-4)$

$C(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4)$ for $k = 0,1,...,293$; $u(k) = 0$ for $k < 0$

The code is punctured in such a way that the following 132 coded bits:

{C(18*j+1), C(18*j+6), C(18*j+11), C(18*j+15) for j = 0,1,...,31} and the bits C(577), C(582), C(584) and C(587) are not transmitted.

The result is a block of 456 coded bits, {c(0),c(1),..., c(455)}

3.8.4 Interleaving

The interleaving is done as specified for the TCH/F9.6 in section 3.3.4

3.8.5 Mapping on a Burst

The mapping is done as specified for TCH/FS in section 3.1.4. On bitstealing by a FACCH, see section 4.2.5.

4 Control Channels

4.1 Slow associated control channel (SACCH)

4.1.1 Block constitution

The message delivered to the encoder has a fixed size of 184 information bits {d(0),d(1),...,d(183)}. It is delivered on a burst mode.

4.1.2 Block code

a) Parity bits:

The block of 184 information bits is protected by 40 extra bits used for error correction and detection. These bits are added to the 184 bits according to a shortened binary cyclic code (FIRE code) using the generator polynomial:

$g(D) = (D^{23} + 1)*(D^{17} + D^3 + 1)$

The encoding of the cyclic code is performed in a systematic form, which means that, in GF(2), the polynomial:

$d(0)D^{223} + d(1)D^{222} +...+d(183)D^{40} + p(1)D^{38} +...+p(38)D + p(39)$

where {p(0),p(1),...,p(39)} are the parity bits , when divided by g(D) yields a remainder equal to:

$1 + D + D^2 +...+ D^{39}$.

b) Tail bits

Four tail bits equal to 0 are added to the information and parity bits, the result being a block of 228 bits.

$u(k) = d(k)$ for k= 0,1,...,183

$u(k) = p(k-184)$ for k = 184,185,...,223

$u(k) = 0$ for k = 224,225,226,227 (tail bits)

4.1.3 Convolutional encoder

This block of 228 bits is encoded with the 1/2 rate convolutional code (identical to the one used for TCH/FS) defined by the polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

This results in a block of 456 coded bits: {c(0),c(1),...,c(455)} defined by:

$c(2k) = u(k) + u(k-3) + u(k-4)$

$c(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4)$ for k = 0,1,...,227 ; u(k) = 0 for k < 0

4.1.4 Interleaving

The coded bits are reordered and interleaved according to the following rule:

$i(B,j) = c(n,k)$ for $k = 0,1,...,455$

$n = 0,1,...,N,N+1,...$

$B = B_0 + 4n + (k \bmod 4)$

$j = 2((49k) \bmod 57) + ((k \bmod 8) \operatorname{div} 4)$

See table 1. The result of the reordering of bits is the same as given for a TCH/FS (subclause 3.1.3) as can be seen from the evaluation of the bit number-index j, distributing the 456 bits over 4 blocks on even numbered bits and 4 blocks on odd numbered bits. The resulting 4 blocks are built by putting blocks with even numbered bits and blocks with odd numbered bits together into one block.

The block of coded data is interleaved "block rectangular" where a new data block starts every 4th block and is distributed over 4 blocks.

4.1.5 Mapping on a Burst

The mapping is given by the rule:

$e(B,j) = i(B,j)$ and $e(B,59+j) = i(B,57+j)$ for $j = 0,1,...,56$

and

$e(B,57) = hl(B)$ and $e(B,58) = hu(B)$

The two bits labelled hl(B) and hu(B) on burst number B are flags used for indication of control channel signalling. They are set to "1" for a SACCH.

4.2 Fast associated control channel at full rate (FACCH/F)

4.2.1 Block constitution

The message delivered to the encoder has a fixed size of 184 information bits. It is delivered on a burst mode.

4.2.2 Block code

The block encoding is done as specified for the SACCH in subclause 4.1.2.

4.2.3 Convolutional encoder

The convolutional encoding is done as specified for the SACCH in subclause 4.1.3.

4.2.4 Interleaving

The interleaving is done as specified for the TCH/FS in subclause 3.1.3.

4.2.5 Mapping on a Burst

A FACCH/F frame of 456 coded bits is mapped on 8 consecutive bursts as specified for the TCH/FS in subclause 3.1.4. As a FACCH is transmitted on bits which are stolen in a burst from the traffic channel, the even numbered bits in the first 4 bursts and the odd numbered bits of the last 4 bursts are stolen.

To indicate this to the receiving device the flags hl(B) and hu(B) have to be set according to the following rule:

hu(B) = 1 for the first 4 bursts (even numbered bits are stolen);

hl(B) = 1 for the last 4 bursts (odd numbered bits are stolen).

The consequences of this bitstealing by a FACCH/F is for a:

- speech channel (TCH/FS) and data channel (TCH/F2.4):

  One full frame of data is stolen by the FACCH.

- Data channel (TCH/F14.4):

  The bitstealing by a FACCH/F disturbs a maximum of 96 of the 456 coded bits generated from an input data block of 290 bits.

- Data channel (TCH/F9.6):

  The bitstealing by a FACCH/F disturbs a maximum of 96 coded bits generated from an input frame of four data blocks. A maximum of 24 of the 114 coded bits resulting from one input data block of 60 bits may be disturbed.

- Data channel (TCH/F4.8):

  The bit stealing by FACCH/F disturbs a maximum of 96 coded bits generated from an input frame of two data blocks. A maximum of 48 of the 228 coded bits resulting from one input data block of 60 bits may be disturbed.

NOTE: In the case of consecutive stolen frames, a number of bursts will have both the even and the odd bits stolen and both flags hu(B) and hl(B) must be set to 1.

4.3 Fast associated control channel at half rate (FACCH/H)

4.3.1 Block constitution

The message delivered to the encoder has a fixed size of 184 information bits. It is delivered on a burst mode.

4.3.2 Block code

The block encoding is done as specified for the SACCH in subclause 4.1.2.

4.3.3 Convolutional encoder

The convolutional encoding is done as specified for the SACCH in subclause 4.1.3.

4.3.4 Interleaving

The coded bits are reordered and interleaved according to the following rule:

$i(B,j) = c(n,k)$ for $k = 0,1,\ldots,455$

$n = 0,1,\ldots,N,N+1,\ldots$

$B = B_0 + 4n + (k \bmod 8) - 4((k \bmod 8) \operatorname{div} 6)$

$j = 2((49k) \bmod 57) + ((k \bmod 8) \operatorname{div} 4)$

See table 1. The result of the reordering of bits is the same as given for a TCH/FS (subclause 3.1.3) as can be seen from the evaluation of the bit number-index j, distributing the 456 bits over 4 blocks on even numbered bits and 4 blocks on odd

numbered bits. The 2 last blocks with even numbered bits and the 2 last blocks with odd numbered bits are put together into 2 full middle blocks.

The block of coded data is interleaved "block diagonal" where a new data block starts every 4th block and is distributed over 6 blocks.

4.3.5 Mapping on a Burst

A FACCH/H frame of 456 coded bits is mapped on 6 consecutive bursts by the rule:

$e(B,j) = i(B,j)$ and $e(B,59+j) = i(B,57+j)$ for $j = 0,1,...,56$

and

$e(B,57) = hl(B)$ and $e(B,58) = hu(B)$

As a FACCH/H is transmitted on bits which are stolen from the traffic channel, the even numbered bits of the first 2 bursts, all bits of the middle 2 bursts and the odd numbered bits of the last 2 bursts are stolen.

To indicate this to the receiving device the flags hl(B) and hu(B) have to be set according to the following rule:

$hu(B) = 1$ for the first 2 bursts (even numbered bits are stolen)

$hu(B) = 1$ and $hl(B) = 1$ for the middle 2 bursts (all bits are stolen)

$hl(B) = 1$ for the last 2 bursts (odd numbered bits are stolen)

The consequences of this bitstealing by a FACCH/H is for a:

- speech channel (TCH/HS):

  two full consecutive speech frames are stolen by a FACCH/H.

- data channel (TCH/H4.8):

  The bitstealing by FACCH/H disturbs a maximum of 96 coded bits generated from an input frame of four data blocks. A maximum of 24 out of the 114 coded bits resulting from one input data block of 60 bits may be disturbed.

- data channel (TCH/H2.4):

  The bitstealing by FACCH/H disturbs a maximum of 96 coded bits generated from an input frame of four data blocks. A maximum of 24 out of the 114 coded bits resulting from one input data block of 36 bits may be disturbed.

NOTE: In the case of consecutive stolen frames, two overlapping bursts will have both the even and the odd numbered bits stolen and both flags hu(B) and hl(B) must be set to 1.

4.4 Broadcast control, Paging, Access grant, Notification and Cell broadcast channels (BCCH, PCH, AGCH, NCH, CBCH)

The coding scheme used for the broadcast control , paging, access grant, notification and cell broadcast messages is the same as for the SACCH messages, specified in subclause 4.1.

4.5 Stand-alone dedicated control channel (SDCCH)

The coding scheme used for the dedicated control channel messages is the same as for SACCH messages, specified in subclause 4.1.

4.6 Random access channel (RACH)

The burst carrying the random access uplink message has a different structure. It contains 8 information bits d(0),d(1),...,d(7).

Six parity bits p(0),p(1),...,p(5) are defined in such a way that in GF(2) the binary polynomial:

d(0)D^{13} +...+ d(7)D^6 + p(0)D^5 +...+ p(5), when divided by $D^6 + D^5 + D^3 + D^2 + D + 1$ yields a remainder equal to $D^5 + D^4 + D^3 + D^2 + D + 1$.

The six bits of the BSIC, {B(0),B(1),...,B(5)}, of the BS to which the Random Access is intended, are added bitwise modulo 2 to the six parity bits, {p(0),p(1),...,p(5)}. This results in six colour bits, C(0) to C(5) defined as C(k) = b(k) + p(k) (k = 0 to 5) where:

b(0) = MSB of PLMN colour code

b(5) = LSB of BS colour code.

This defines {u(0),u(1),..., u(17)} by:

u(k) = d(k) for k = 0,1,...,7

u(k) = C(k-8) for k = 8,9,...,13

u(k) = 0 for k = 14,15,16,17 (tail bits)

The bits {e(0),e(1),..., e(35)} are obtained by the same convolutional code of rate 1/2 as for TCH/FS, defined by the polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

and with:

e(2k) = u(k) + u(k-3) + u(k-4)

e(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4) for k = 0,1,...,17 ; u(k) = 0 for k < 0

4.7 Synchronization channel (SCH)

The burst carrying the synchronization information on the downlink BCCH has a different structure. It contains 25 information bits {d(0),d(1),..., d(24)}, 10 parity bits {p(0),p(1),..., p(9)} and 4 tail bits. The precise ordering of the information bits is given in GSM 04.08.

The ten parity bits {p(0),p(1),,...,p(9)} are defined in such a way that in GF(2) the binary polynomial:

d(0)D^{34} +...+ d(24)D^{10} + p(0)D^9 +...+ p(9), when divided by:

$D^{10} + D^8 + D^6 + D^5 + D^4 + D^2 + 1$, yields a remainder equal to:

$D^9 + D^8 + D^7 + D^6 + D^5 + D^4 + D^3 + D^2 + D + 1$.

Thus the encoded bits {u(0),u(1),...,u(38)} are:

u(k) = d(k) for k = 0,1,...,24

u(k) = p(k-25) for k = 25,26,...,34

u(k) = 0 for k = 35,36,37,38 (tail bits)

The bits {e(0),e(1),..., e(77)} are obtained by the same convolutional code of rate 1/2 as for TCH/FS, defined by the polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

and with:

$e(2k) = u(k) + u(k-3) + u(k-4)$

$e(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4)$ for $k = 0,1,....,77$; $u(k) = 0$ for $k < 0$

4.8 Access Burst on circuit switched channels other than RACH

The encoding of this burst is as defined in subclause 4.6 for the random access channel (RACH). The BSIC used shall be the BSIC of the BTS to which the burst is intended.

4.9 Access Bursts for uplink access on a channel used for VGCS

The encoding of this burst is as defined in subclause 4.5 for the RACH. The BSIC used by the Mobile Station shall be the BSIC indicated by network signalling, or if not thus provided, the last received BSIC on the SCH of the current cell.

5 Packet Switched Channels

5.1 Packet data traffic channel (PDTCH)

Four coding schemes are specified for the packet data traffic channels. For the three coding schemes CS-2 to CS-4, the first three bits (USF-bits) of the data block are encoded such that the first twelve coded bits are representing the same bit pattern, irrespective of the coding scheme, depending only on the USF-bits. For these coding schemes, the USF-bits can therefore always be decoded from these twelve bits in the same way. It should be noted that the USF precoding is done in the uplink direction for coding schemes CS-2 – CS-4, despite the fact that uplink RLC data block structure (GSM 04.60) does not define USF-field.

5.1.1 Packet data block type 1 (CS-1)

The coding scheme used for packet data block type 1 is the same as for SACCH as specified in section 4.1.

The flags hl(B) and hu(B) set to “1” identify the coding scheme CS-1.

5.1.2 Packet data block type 2 (CS-2)

5.1.2.1 Block constitution

The message delivered to the encoder has a fixed size of 271 information bits {d(0),d(1),...,d(270)}. It is delivered on a burst mode.

5.1.2.2 Block code

a) USF precoding:

The first three bits d(0),d(1),d(2) are precoded into six bits u’(0),u’(1),...,u’(5) according to the following table:

d(0),d(1),d(2)	u'(0),u'(1),...,u'(5)
000	000 000
001	001 011
010	010 110
011	011 101
100	100 101
101	101 110
110	110 011
111	111 000

b) Parity bits:

Sixteen parity bits p(0),p(1),...,p(15) are defined in such a way that in GF(2) the binary polynomial:

$d(0)D^{286} + ... + d(270)D^{16} + p(0)D^{15} + ... + p(15)$, when divided by:

$D^{16} + D^{12} + D^5 + 1$, yields a remainder equal to:

$D^{15} + D^{14} + D^{13} + D^{12} + D^{11} + D^{10} + D^9 + D^8 + D^7 + D^6 + D^5 + D^4 + D^3 + D^2 + D + 1$.

c) Tail bits:

Four tail bits equal to 0 are added to the information and parity bits, the result being a block of 294 bits {u(0),u(1),...,u(293)}:

$u(k) = u'(k)$ for $k = 0,1,...,5$

$u(k) = d(k-3)$ for $k = 6,7,...,273$

$u(k) = p(k-274)$ for $k = 274,275,...,289$

$u(k) = 0$ for $k = 290,291,292,293$ (tail bits)

5.1.2.3 Convolutional encoder

This block of 294 bits {u(0),u(1),...,u(293)} is encoded with the 1/2 rate convolutional code (identical to the one used for TCH/FS) defined by the polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

This results in a block of 588 coded bits: {C(0),C(1),...,C(587)} defined by:

$C(2k) = u(k) + u(k-3) + u(k-4)$

$C(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4)$ for $k = 0,1,...,293$; $u(k) = 0$ for $k < 0$

The code is punctured in such a way that the following coded bits:

{C(3+4j) for j = 3,4,...,146 except for j = 9,21,33,45,57,69,81,93,105,117,129,141} are not transmitted

The result is a block of 456 coded bits, {c(0),c(1),...,c(455)}.

5.1.2.4 Interleaving

The interleaving is done as specified for SACCH in section 4.1.4.

5.1.2.5 Mapping on a burst

The mapping is given by the rule:

$e(B,j) = i(B,j)$ and $e(B,59+j) = i(B,57+j)$ for $j = 0,1,...,56$

and

$e(B+m,57) = q(2m)$ and $e(B+m,58) = q(2m+1)$ for $m = 0,1,2,3$

where

q(0),q(1),...,q(7) = 1,1,0,0,1,0,0,0 identifies the coding scheme CS-2.

5.1.3 Packet data block type 3 (CS-3)

5.1.3.1 Block constitution

The messages delivered to the encoder has a fixed size of 315 information bits {d(0),d(1),...,d(314)}. It is delivered on a burst mode.

5.1.3.2 Block code

a) USF precoding:

The first three bits d(0),d(1),d(2) are precoded into six bits u'(0),u'(1),...,u'(5) as specified for CS-2 in section 5.1.2.2.a).

b) Parity bits:

Sixteen parity bits p(0),p(1),...,p(15) are defined in such a way that in GF(2) the binary polynomial:

$d(0)D^{330} + ... + d(314)D^{16} + p(0)D^{15} + ... + p(15)$, when divided by:

$D^{16} + D^{12} + D^5 + 1$, yields a remainder equal to:

$D^{15} + D^{14} + D^{13} + D^{12} + D^{11} + D^{10} + D^9 + D^8 + D^7 + D^6 + D^5 + D^4 + D^3 + D^2 + D + 1$.

c) Tail bits:

Four tail bits equal to 0 are added to the information and parity bits, the result being a block of 338 bits {u(0),u(1),...,u(337)}:

u(k) = u'(k) for k = 0,1,...,5

u(k) = d(k-3) for k = 6,7,...,317

u(k) = p(k-318) for k = 318,319,...,333

u(k) = 0 for k = 334,335,336,337 (tail bits)

5.1.3.3 Convolutional encoder

This block of 338 bits {u(0),u(1),...,u(337)} is encoded with the 1/2 rate convolutional code (identical to the one used for TCH/FS) defined by the polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

This results in a block of 676 coded bits: {C(0),C(1),...,C(675)} defined by:

C(2k) = u(k) + u(k-3) + u(k-4)

C(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4) for k = 0,1,...,337 ; u(k) = 0 for k < 0

The code is punctured in such a way that the following coded bits:

{C(3+6j) and C(5+6j) for j = 2,3,...,111} are not transmitted

The result is a block of 456 coded bits, {c(0),c(1),...,c(455)}.

5.1.3.4 Interleaving

The interleaving is done as specified for SACCH in subclause 4.1.4.

5.1.3.5 Mapping on a burst

The mapping is given by the rule:

$e(B,j) = i(B,j)$ and $e(B,59+j) = i(B,57+j)$ for $j = 0,1,...,56$

and

$e(B+m,57) = q(2m)$ and $e(B+m,58) = q(2m+1)$ for $m = 0,1,2,3$

where

$q(0),q(1),...,q(7) = 0,0,1,0,0,0,0,1$ identifies the coding scheme CS-3.

5.1.4 Packet data block type 4 (CS-4)

5.1.4.1 Block constitution

The message delivered to the encoder has a fixed size of 431 information bits {d(0),d(1),...,d(430)}. It is delivered on a burst mode.

5.1.4.2 Block code

a) USF precoding:

The first three bits d(0),d(1),d(2) are block coded into twelve bits u'(0),u'(1),...,u'(11) according to the following table:

d(0),d(1),d(2)	u'(0),u'(1),...,u'(11)
000	000 000 000 000
001	000 011 011 101
010	001 101 110 110
011	001 110 101 011
100	110 100 001 011
101	110 111 010 110
110	111 001 111 101
111	111 010 100 000

b) Parity bits:

Sixteen parity bits p(0),p(1),...,p(15) are defined in such a way that in GF(2) the binary polynomial:

$d(0)D^{446} + ... + d(430)D^{16} + p(0)D^{15} + ... + p(15)$, when divided by:

$D^{16} + D^{12} + D^5 + 1$, yields a remainder equal to:

$D^{15} + D^{14} + D^{13} + D^{12} + D^{11} + D^{10} + D^9 + D^8 + D^7 + D^6 + D^5 + D^4 + D^3 + D^2 + D + 1$.

The result is a block of 456 coded bits, {c(0),c(1),...,c(455)}:

$c(k) = u'(k)$ for $k = 0,1,...,11$

$c(k) = d(k-9)$ for $k = 12,13,...,439$

$c(k) = p(k-440)$ for $k = 440,441,...,455$

5.1.4.3 Convolutional encoder

No convolutional coding is done.

5.1.4.4 Interleaving

The interleaving is done as specified for SACCH in section 4.1.4.

5.1.4.5 Mapping on a burst

The mapping is given by the rule:

$e(B,j) = i(B,j)$ and $e(B,59+j) = i(B,57+j)$ for $j = 0,1,...,56$

and

$e(B+m,57) = q(2m)$ and $e(B+m,58) = q(2m+1)$ for $m = 0,1,2,3$

where

$q(0),q(1),...,q(7) = 0,0,0,1,0,1,1,0$ identifies the coding scheme CS-4.

5.2 Packet control channels (PACCH, PBCCH, PAGCH, PPCH, PNCH, PTCCH)

The coding scheme used for PACCH, PBCCH, PAGCH, PPCH, PNCH and downlink PTCCH is the same as for SACCH as specified in section 4.1.

The coding scheme used for uplink PTCCH is the same as for PRACH as specified in section 5.3.

5.3 Packet random access channel (PRACH)

Two coding schemes are specified for access bursts on the packet switched channels. The packet access burst containing 8 information bits and the extended packet access burst containing 11 information bits.

5.3.1 Packet Access Burst

The encoding of this burst is as defined in section 4.6 for the random access channel (RACH). The BSIC used shall be the BSIC of the BTS to which the burst is intended.

5.3.2 Extended Packet Access Burst

The burst carrying the extended packet random access uplink message contains 11 information bits d(0),d(1),...,d(10).

Six parity bits p(0),p(1),...,p(5) are defined in such a way that in GF(2) the binary polynomial:

$d(0)D^{16} +...+ d(10)D^6 + p(0)D^5 +...+ p(5)$, when divided by $D^6 + D^5 + D^3 + D^2 + D + 1$ yields a remainder equal to $D^5 + D^4 + D^3 + D^2 + D + 1$.

The six bits of the BSIC, {B(0),B(1),...,B(5)}, of the BTS to which the Random Access is intended, are added bitwise modulo 2 to the six parity bits, {p(0),p(1),...,p(5)}. This results in six colour bits, C(0) to C(5) defined as $C(k) = b(k) + p(k)$ ($k = 0$ to 5) where:

b(0) = MSB of PLMN colour code

b(5) = LSB of BS colour code.

This defines {u(0),u(1),..., u(20)} by:

$u(k) = d(k)$ for $k = 0,1,...,10$

$u(k) = C(k-11)$ for $k = 11,12,...,16$

$u(k) = 0$ for $k = 17,18,19,20$ (tail bits)

The coded bits {c(0),c(1),..., c(41)} are obtained by the same convolutional code of rate 1/2 as for TCH/FS, defined by the polynomials:

$G0 = 1 + D^3 + D^4$

$G1 = 1 + D + D^3 + D^4$

and with:

$c(2k) = u(k) + u(k-3) + u(k-4)$

$c(2k+1) = u(k) + u(k-1) + u(k-3) + u(k-4)$ for $k = 0,1,...,20$; $u(k) = 0$ for $k < 0$

The code is punctured in such a way that the following coded bits:

c(0), c(2), c(5), c(37), c(39), c(41) are not transmitted.

This results in a block of 36 coded bits, {e(0), e(1),...,e(35)}.

5.4 Access Burst on packet switched channels other than PRACH

The encoding of this burst is as defined in section 5.3 for the packet random access channel (PRACH). The BSIC used shall be the BSIC of the BTS to which the burst is intended.

Table 1: Reordering and partitioning of a coded block of 456 bits into 8 sub-blocks

k mod 8=	0	1	2	3
j=0	k=0	57	114	171
2	64	121	178	235
4	128	185	242	299
6	192	249	306	363
8	256	313	370	427
10	320	377	434	35
	384	441	42	99
	448	49	106	163
	56	113	170	227
	120	177	234	291
20	184	241	298	355
	248	305	362	419
	312	369	426	27
	376	433	34	91
	440	41	98	155
30	48	105	162	219
	112	169	226	283
	176	233	290	347
	240	297	354	411
	304	361	418	19
40	368	425	26	83
	432	33	90	147
	40	97	154	211
	104	161	218	275
	168	225	282	339
50	232	289	346	403
	296	353	410	11
	360	417	18	75
	424	25	82	139
	32	89	146	203
60	96	153	210	267
	160	217	274	331
	224	281	338	395
	288	345	402	3
	352	409	10	67
70	416	17	74	131
	24	81	138	195
	88	145	202	259
	152	209	266	323
	216	273	330	387
80	280	337	394	451
	344	401	2	59
	408	9	66	123
	16	73	130	187
	80	137	194	251
90	144	201	258	315
	208	265	322	379
	272	329	386	443
	336	393	450	51
	400	1	58	115
100	8	65	122	179
	72	129	186	243
	136	193	250	307
	200	257	314	371
	264	321	378	435
110	328	385	442	43
112	392	449	50	107

k mod 8=	4	5	6	7
j=1	228	285	342	399
3	292	349	406	7
5	356	413	14	71
7	420	21	78	135
9	28	85	142	199
11	92	149	206	263
	156	213	270	327
	220	277	334	391
	284	341	398	455
	348	405	6	63
21	412	13	70	127
	20	77	134	191
	84	141	198	255
	148	205	262	319
	212	269	326	383
31	276	333	390	447
	340	397	454	55
	404	5	62	119
	12	69	126	183
	76	133	190	247
41	140	197	254	311
	204	261	318	375
	268	325	382	439
	332	389	446	47
	396	453	54	111
51	4	61	118	175
	68	125	182	239
	132	189	246	303
	196	253	310	367
	260	317	374	431
61	324	381	438	39
	388	445	46	103
	452	53	110	167
	60	117	174	231
	124	181	238	295
71	188	245	302	359
	252	309	366	423
	316	373	430	31
	380	437	38	95
	444	45	102	159
81	52	109	166	223
	116	173	230	287
	180	237	294	351
	244	301	358	415
	308	365	422	23
91	372	429	30	87
	436	37	94	151
	44	101	158	215
	108	165	222	279
	172	229	286	343
101	236	293	350	407
	300	357	414	15
	364	421	22	79
	428	29	86	143
	36	93	150	207
111	100	157	214	271
113	164	221	278	335

Table 2: Subjective importance of encoded bits for the full rate speech TCH (Parameter names and bit indices refer to GSM 06.10)

Importance class	Parameter name	Parameter number	Bit index	Label	Class
1	Log area ratio 1	1	5	d0	
	block amplitude	12,29,46,63	5	d1, d2, d3, d4	
2	Log area ratio 1	1	4		
	Log area ratio 2	2	5		
	Log area ratio 3	3	4		
3	Log area ratio 1	1	3		
	Log area ratio 2	2	4		
	Log area ratio 3	3	3		
	Log area ratio 4	4	4		
	LPT lag	9,26,43,60	6		1
	block amplitude	12,29,43,63	4		with
	Log area ratio 2,5,6	2,5,6	3		parity
	LPT lag	9,26,43,60	5		check
	LPT lag	9,26,43,60	4		
	LPT lag	9,26,43,60	3		
	LPT lag	9,26,43,60	2		
4	block amplitude	12,29,43,63	3		
	Log area ratio 1	1	2		
	Log area ratio 4	4	3		
	Log area ratio 7	7	2		
	LPT lag	9,26,43,60	1	...d48, d49	
	Log area ratio 5,6	5,6	2	d50	
	LPT gain	10,27,44,61	1		
	LPT lag	9,26,43,60	0		
	Grid position	11,28,45,62	1		
5	Log area ratio 1	1	1		
	Log area ratio 2,3,8,4	2,3,8,4	2		
	Log area ratio 5,7	5,7	1		
	LPT gain	10,27,44,61	0		
	block amplitude	12,29,43,63	2		1
	RPE pulses	13..25	2		with
	RPE pulses	30..42	2		parity
	RPE pulses	47..59	2		check
	RPE pulses	64..76	2		
	Grid position	11,28,45,62	0		
	block amplitude	12,29,43,63	1		
	RPE pulses	13..25	1		
	RPE pulses	30..42	1		
	RPE pulses	47..59	1		
	RPE pulses	64..67	1	...d181	
	RPE pulses	68..76	1	d182	
6	Log area ratio 1	1	0		
	Log area ratio 2,3,6	2,3,6	1		
	Log area ratio 7	7	0		
	Log area ratio 8	8	1		
	Log area ratio 8,3	8,3	0		
	Log area ratio 4	4	1		2
	Log area ratio 4,5	4,5	0		
	block amplitude	12,29,43,63	0		
	RPE pulses	13..25	0		
	RPE pulses	30..42	0		
	RPE pulses	47..59	0		
	RPE pulses	64..67	0		
	Log area ratio 2,6	2,6	0	...d259	

Table 3a: Subjective importance of encoded bits for the half rate speech TCH for unvoiced speech frames (Parameter names and bit indices refer to GSM 06.20)

Parameter name	Bit index	Label	Class
R0	1	d0	
LPC 3	7	d1	
GSP 0-1	2	d2	
GSP 0-2	2	d3	
GSP 0-3	2	d4	
GSP 0-4	2	d5	
LPC 1	0	d6	
LPC 2	5...1	d7...d11	
LPC 3	6...1	d12...	
Code 1-2	0		
Code 2-2	6...0		
Code 1-3	6...0		1
Code 2-3	6...3		
LPC3	0		without
R0	0		parity
INT-LPC	0		check
Code 1-2	1...6		
Code 2-1	0...6		
Code 1-1	0...6		
GSP 0-4	0		
GSP 0-3	0		
GSP 0-2	0		
GSP 0-1	0		
LPC 2	0		
GSP 0-4	1		
GSP 0-3	1		
GSP 0-2	1		
GSP 0-1	1		
LPC 1	1...4	...d72	
LPC 1	5	d73...	
GSP 0-4	3		
GSP 0-3	3		
GSP 0-2	3		
GSP 0-1	3		
LPC2	6...8		1
GSP 0-4	4		
GSP 0-3	4		with
GSP 0-2	4		parity
GSP 0-1	4		check
LPC 1	6...9		
R0	2		
LPC 1	10		
R0	3,4		
Mode	0,1	...d94	
Code 2-4	0...6	d95...	
Code 1-4	0...6		2
Code 2-3	0...2	...d111	

Table 3b: Subjective importance of encoded bits for the half rate speech TCH for voiced speech frames (Parameter names and bit indices refer to GSM 06.20)

Parameter name	Bit index	Label	Class
LPC 1	2,1	d0, d1	
LPC 2	6...4	d2...	
GSP 0-1	4		
GSP 0-2	4		
GSP 0-3	4		
GSP 0-4	4		
GSP 0-1	3		
GSP 0-2	3		
GSP 0-3	3		
GSP 0-4	3		
GSP 0-1	2		
GSP 0-2	2		
GSP 0-3	2		
GSP 0-4	2		
Code 1	8...0		
Code 2	8...5		
Code 2	2...0		
Code 3	8		
Code 2	4,3		
GSP 0-1	1		
GSP 0-2	1		
GSP 0-3	1		
GSP 0-4	1		1
GSP 0-1	0		
GSP 0-2	0		without
GSP 0-3	0		parity
GSP 0-4	0		check
INT-LPC	0		
LPC 2	0		
LPC 3	0		
LAG 4	0		
LPC 3	1		
LPC 2	1		
LAG 4	1		
LAG 3	0		
LAG 2	0		
LAG 1	0		
LAG 4	2		
LAG 3	1		
LAG 2	1		
LAG 1	1		
LPC 3	2...4		
LPC 2	2		
LPC 3	5,6		
LPC 2	3		
R0	0		
LPC 3	7		
LPC 1	0		
LAG 4	3		
LAG 3	2		
LAG 2	2		
LAG 1	2		
R0	1	...d72	

Parameter name	Bit index	Label	Class
LAG 3	3	d73...	
LAG 2	3		
LAG 1	3,4		1
LPC 2	7,8		
LPC 1	3...6		with
R0	2		parity
LAG 1	5...7		check
LPC 1	7...10		
R0	3,4		
Mode	0,1	...d94	
Code 4	0...8	d95...	2
Code 3	0...7	...d111	

Table 4: Reordering and partitioning of a coded block of 228 bits into 4 sub-blocks for TCH/HS

b=	0	1
i=0	k=0	150
2	38	188
4	76	226
6	114	14
8	152	52
10	190	90
	18	128
	56	166
	94	204
	132	32
20	170	70
	208	108
	8	146
	46	184
	84	222
30	122	10
	160	48
	198	86
	28	124
	66	162
40	104	200
	142	30
	180	68
	218	106
	4	144
50	42	182
	80	220
	118	6
	156	44
	194	82
60	22	120
	60	158
	98	196
	136	24
	174	62
70	212	100
	12	138
	50	176
	88	214
	126	2
80	164	40
	202	78
	34	116
	72	154
	110	192
90	148	26
	186	64
	224	102
	16	140
	54	178
100	92	216
	130	20
	168	58
	206	96
	36	134
110	74	172
112	112	210

b=	2	3
i=1	k=1	151
3	39	189
5	77	227
7	115	15
9	153	53
11	191	91
	19	129
	57	167
	95	205
	133	33
21	171	71
	209	109
	9	147
	47	185
	85	223
31	123	11
	161	49
	199	87
	29	125
	67	163
41	105	201
	143	31
	181	69
	219	107
	5	145
51	43	183
	81	221
	119	7
	157	45
	195	83
61	23	121
	61	159
	99	197
	137	25
	175	63
71	213	101
	13	139
	51	177
	89	215
	127	3
81	165	41
	203	79
	35	117
	73	155
	111	193
91	149	27
	187	65
	225	103
	17	141
	55	179
101	93	217
	131	21
	169	59
	207	97
	37	135
111	75	173
113	113	2
		11

Table 5: Enhanced Full rate Source Encoder output parameters in order of occurrence and bit allocation within the speech frame of 244 bits/20 ms(Parameter names and bit indices refer to GSM 06.60)

Bits (MSB-LSB)	Description
s1 - s7	index of 1st LSF submatrix
s8 - s15	index of 2nd LSF submatrix
s16 - s23	index of 3rd LSF submatrix
s24	sign of 3rd LSF submatrix
s25 - s32	index of 4th LSF submatrix
s33 - s38	index of 5th LSF submatrix
subframe 1	
s39 - s47	adaptive codebook index
s48 - s51	adaptive codebook gain
s52	sign information for 1st and 6th pulses
s53 - s55	position of 1st pulse
s56	sign information for 2nd and 7th pulses
s57 - s59	position of 2nd pulse
s60	sign information for 3rd and 8th pulses
s61 - s63	position of 3rd pulse
s64	sign information for 4th and 9th pulses
s65 - s67	position of 4th pulse
s68	sign information for 5th and 10th pulses
s69 - s71	position of 5th pulse
s72 - s74	position of 6th pulse
s75 - s77	position of 7th pulse
s78 - s80	position of 8th pulse
s81 - s83	position of 9th pulse
s84 - s86	position of 10th pulse
s87 - s91	fixed codebook gain
subframe 2	
s92 - s97	adaptive codebook index (relative)
s98 - s141	same description as s48 - s91
subframe 3	
s142 - s194	same description as s39 - s91
subframe 4	
s195 - s244	same description as s92 - s141

Table 6: Ordering of enhanced full rate speech parameters for the channel encoder (subjective importance of encoded bits) (after preliminary channel coding) (Parameter names refers to GSM 06.60)

Description	Bits (Table 5)	Bit index within parameter
CLASS 1a: 50 bits (protected by 3 bit TCH-FS CRC)		
LTP-LAG 1	w39 - w44	b8, b7, b6, b5, b4, b3
LTP-LAG 3	w146 - w151	b8, b7, b6, b5, b4, b3
LTP-LAG 2	w94 - w95	b5, b4
LTP-LAG 4	w201 - w202	b5, b4
LTP-GAIN 1	n48	b3
FCB-GAIN 1	w89	b4
LTP-GAIN 2	w100	b3
FCB-GAIN 2	w141	b4
LTP-LAG 1	w45	b2
LTP-LAG 3	w152	b2
LTP-LAG 2	w96	b3
LTP-LAG 4	w203	b3
LPC 1	w2 - w3	b5, b4
LPC 2	w8	b7
LPC 2	w10	b5
LPC 3	w18 - w19	b6, b5
LPC 3	w24	b0
LTP-LAG 1	w46 - w47	b1, b0
LTP-LAG 3	w153 - w154	b1, b0
LTP-LAG 2	w97	b2
LTP-LAG 4	w204	b2
LPC 1	w4 - w5	b3, b2
LPC 2	w11 - w12	b4, b3
LPC 3	w16	b8
LPC 2	w9	b6
LPC 1	w6 - w7	b1, b0
LPC 2	w13	b2
LPC 3	w17	b7
LPC 3	w20	b4
LTP-LAG 2	w98	b1
LTP-LAG 4	w205	b1
CLASS 1b: 132 bits (protected)		
LPC 1	w1	b6
LPC 2	w14 - w15	b1, b0
LPC 3	w21	b3
LPC 4	w25 - w26	b7, b6
LPC 4	w28	b4
LTP-GAIN 3	w155	b3
LTP-GAIN 4	w207	b3
FCB-GAIN 3	w196	b4
FCB-GAIN 4	w248	b4
FCB-GAIN 1	w90	b3
FCB-GAIN 2	w142	b3
FCB-GAIN 3	w197	b3
FCB-GAIN 4	w249	b3
CRC-POLY	w253 - w260	b7, b6, b5, b4, b3, b2, b1, b0
LTP-GAIN 1	w49	b2

(continued)

Table 6 (continued): Ordering of enhanced full rate speech parameters for the channel encoder (subjective importance of encoded bits) (after preliminary channel coding) (Parameter names refers to GSM 06.60)

Description	Bits (Table 5)	Bit index within parameter
LTP-GAIN 2	w101	b2
LTP-GAIN 3	w156	b2
LTP-GAIN 4	w208	b2
LPC 3	w22 - w23	b2, b1
LPC 4	w27	b5
LPC 4	w29	b3
PULSE 1_1	w52	b3
PULSE 1_2	w56	b3
PULSE 1_3	w60	b3
PULSE 1_4	w64	b3
PULSE 1_5	w68	b3
PULSE 2_1	w104	b3
PULSE 2_2	w108	b3
PULSE 2_3	w112	b3
PULSE 2_4	w116	b3
PULSE 2_5	w120	b3
PULSE 3_1	w159	b3
PULSE 3_2	w163	b3
PULSE 3_3	w167	b3
PULSE 3_4	w171	b3
PULSE 3_5	w175	b3
PULSE 4_1	w211	b3
PULSE 4_2	w215	b3
PULSE 4_3	w219	b3
PULSE 4_4	w223	b3
PULSE 4_5	w227	b3
FCB-GAIN 1	w91	b2
FCB-GAIN 2	w143	b2
FCB-GAIN 3	w198	b2
FCB-GAIN 4	w250	b2
LTP-GAIN 1	w50	b1
LTP-GAIN 2	w102	b1
LTP-GAIN 3	w157	b1
LTP-GAIN 4	w209	b1
LPC 4	w30 - w32	b2, b1, b0
LPC 5	w33 - w36	b5, b4, b3, b2
LTP-LAG 2	w99	b0
LTP-LAG 4	w206	b0
PULSE 1_1	w53	b2
PULSE 1_2	w57	b2

(continued)

Table 6 (continued): Ordering of enhanced full rate speech parameters for the channel encoder (subjective importance of encoded bits) (after preliminary channel coding) (Parameter names refers to GSM 06.60)

Description	Bits (Table 5)	Bit index within parameter
PULSE 1_3	w61	b2
PULSE 1_4	w65	b2
PULSE 1_5	w69	b2
PULSE 2_1	w105	b2
PULSE 2_2	w109	b2
PULSE 2_3	w113	b2
PULSE 2_4	w117	b2
PULSE 2_5	w121	b2
PULSE 3_1	w160	b2
PULSE 3_2	w164	b2
PULSE 3_3	w168	b2
PULSE 3_4	w172	b2
PULSE 3_5	w176	b2
PULSE 4_1	w212	b2
PULSE 4_2	w216	b2
PULSE 4_3	w220	b2
PULSE 4_4	w224	b2
PULSE 4_5	w228	b2
PULSE 1_1	w54	b1
PULSE 1_2	w58	b1
PULSE 1_3	w62	b1
PULSE 1_4	w66	b1
PULSE 2_1	w106	b1
PULSE 2_2	w110	b1
PULSE 2_3	w114	b1
PULSE 2_4	w118	b1
PULSE 3_1	w161	b1
PULSE 3_2	w165	b1
PULSE 3_3	w169	b1
PULSE 3_4	w173	b1
PULSE 4_1	w213	b1
PULSE 4_3	w221	b1
PULSE 4_4	w225	b1
FCB-GAIN 1	w92	b1
FCB-GAIN 2	w144	b1
FCB-GAIN 3	s199	b1
FCB-GAIN 4	w251	b1
LTP-GAIN 1	w51	b0
LTP-GAIN 2	w103	b0
LTP-GAIN 3	w158	b0
LTP-GAIN 4	w210	b0
FCB-GAIN 1	w93	b0
FCB-GAIN 2	w145	b0
FCB-GAIN 3	w200	b0
FCB-GAIN 4	w252	b0
PULSE 1_1	w55	b0
PULSE 1_2	w59	b0
PULSE 1_3	w63	b0
PULSE 1_4	w67	b0
PULSE 2_1	w107	b0
PULSE 2_2	w111	b0
PULSE 2_3	w115	b0
PULSE 2_4	w119	b0
PULSE 3_1	w162	b0
PULSE 3_2	w166	b0
PULSE 3_3	w170	b0

(continued)

Table 6 (continued): Ordering of enhanced full rate speech parameters for the channel encoder (subjective importance of encoded bits) (after preliminary channel coding) (Parameter names refers to GSM 06.60)

Description	Bits (Table 5)	Bit index within parameter
PULSE 3_4	w174	b0
PULSE 4_1	w214	b0
PULSE 4_3	w222	b0
PULSE 4_4	w226	b0
LPC 5	w37 - w38	b1, b0
CLASS 2: 78 bits (unprotected)		
PULSE 1_5	w70	b1
PULSE 1_5	w72 - w73	b1, b1
PULSE 2_5	w122	b1
PULSE 2_5	w124 - s125	b1, b1
PULSE 3_5	w177	b1
PULSE 3_5	w179 - w180	b1, b1
PULSE 4_5	w229	b1
PULSE 4_5	w231 - w232	b1, b1
PULSE 4_2	w217 - w218	b1, b0
PULSE 1_5	w71	b0
PULSE 2_5	w123	b0
PULSE 3_5	w178	b0
PULSE 4_5	w230	b0
PULSE 1_6	w74	b2
PULSE 1_7	w77	b2
PULSE 1_8	w80	b2
PULSE 1_9	w83	b2
PULSE 1_10	w86	b2
PULSE 2_6	w126	b2
PULSE 2_7	w129	b2
PULSE 2_8	w132	b2
PULSE 2_9	w135	b2
PULSE 2_10	w138	b2
PULSE 3_6	w181	b2
PULSE 3_7	w184	b2
PULSE 3_8	w187	b2
PULSE 3_9	w190	b2
PULSE 3_10	w193	b2
PULSE 4_6	w233	b2
PULSE 4_7	w236	b2
PULSE 4_8	w239	b2
PULSE 4_9	w242	b2
PULSE 4_10	w245	b2
PULSE 1_6	w75	b1
PULSE 1_7	w78	b1
PULSE 1_8	w81	b1
PULSE 1_9	w84	b1
PULSE 1_10	w87	b1
PULSE 2_6	w127	b1
PULSE 2_7	w130	b1
PULSE 2_8	w133	b1
PULSE 2_9	w136	b1
PULSE 2_10	w139	b1
PULSE 3_6	w182	b1
PULSE 3_7	w185	b1
PULSE 3_8	w188	b1
PULSE 3_9	w191	b1
PULSE 3_10	w194	b1
PULSE 4_6	w234	b1
PULSE 4_7	w237	b1

(continued)

Table 6 (concluded): Ordering of enhanced full rate speech parameters for the channel encoder (subjective importance of encoded bits) (after preliminary channel coding) (Parameter names refers to GSM 06.60)

Description	Bits (Table 5)	Bit index within parameter
PULSE 4_8	w240	b1
PULSE 4_9	w243	b1
PULSE 4_10	w246	b1
PULSE 1_6	w76	b0
PULSE 1_7	w79	b0
PULSE 1_8	w82	b0
PULSE 1_9	w85	b0
PULSE 1_10	w88	b0
PULSE 2_6	w128	b0
PULSE 2_7	w131	b0
PULSE 2_8	w134	b0
PULSE 2_9	w137	b0
PULSE 2_10	w140	b0
PULSE 3_6	w183	b0
PULSE 3_7	w186	b0
PULSE 3_8	w189	b0
PULSE 3_9	w192	b0
PULSE 3_10	w195	b0
PULSE 4_6	w235	b0
PULSE 4_7	w238	b0
PULSE 4_8	w241	b0
PULSE 4_9	w244	b0
PULSE 4_10	w247	b0

Annex A (informative): Summary of Channel Types

TCH/EFS:	enhanced full rate speech traffic channel
TCH/FS:	full rate speech traffic channel
TCH/HS:	half rate speech traffic channel
TCH/F14.4	14.4 kbit/s full rate data traffic channel
TCH/F9.6:	9.6 kbit/s full rate data traffic channel
TCH/F4.8:	4.8 kbit/s full rate data traffic channel
TCH/H4.8:	4.8 kbit/s half rate data traffic channel
TCH/F2.4:	2.4 kbit/s full rate data traffic channel
TCH/H2.4:	2.4 kbit/s half rate data traffic channel
SACCH:	slow associated control channel
FACCH/F:	fast associated control channel at full rate
FACCH/H:	fast associated control channel at half rate
SDCCH:	stand-alone dedicated control channel
BCCH:	broadcast control channel
PCH:	paging channel
AGCH	access grant channel
RACH:	random access channel
SCH:	synchronization channel
CBCH:	cell broadcast channel
PDTCH	packet data traffic channel
PACCH	packet associated control channel
PBCCH	packet broadcast control channel
PAGCH	packet access grant channel
PPCH	packet paging channel
PNCH	packet notification channel
PTCCH	packet timing advance control channel
PRACH	packet random access channel

Annex B (informative): Summary of Polynomials Used for Convolutional Codes

$G0 = 1 + D^3 + D^4$	TCH/FS, TCH/EFS, TCH/F14.4, TCH/F9.6, TCH/H4.8, SDCCH, BCCH, PCH, SACCH, FACCH, AGCH, RACH, SCH, PDTCH (CS-1, CS-2, CS3, CS-4), PACCH,PBCCH, PAGCH, PPCH, PNCH, PTCCH, PRACH
$G1 = 1 + D + D^3 + D^4$	TCH/FS, TCH/EFS, TCH/F14.4, TCH/F9.6, TCH/H4.8, SACCH, FACCH, SDCCH, BCCH,PCH, AGCH, RACH, SCH, TCH/F4.8, TCH/F2.4, TCH/H2.4,PDTCH(CS-1, CS-2, CS-3, CS-4), PACCH, PBCCH, PAGCH, PPCH, PNCH, PTCCH, PRACH
$G2 = 1 + D^2 + D^4$	TCH/F4.8, TCH/F2.4, TCH/H2.4
$G3 = 1 + D + D^2 + D^3 + D^4$	TCH/F4.8, TCH/F2.4, TCH/H2.4
$G4 = 1 + D^2 + D^3 + D^5 + D^6$	TCH/HS
$G5 = 1 + D + D^4 + D^6$	TCH/HS
$G6 = 1 + D + D^2 + D^3 + D^4 + D^6$	TCH/HS

Annex C (informative): Change Request History

SPEC	SMG	CR	PHA	VERS	NEW_VE	SUBJECT
05.03	s25	A015	R97	6.0.0	6.1.0	14.4kbps Data Service
05.03	s27		R97	6.1.0	6.1.2	Change of status to EN
05.03	s28	A017	R97	6.1.2	6.2.0	Clarification on the definition of USF precoding

History

Document history				
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